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Exploring Validity Evidence for Mathematics Self-Concept of Adolescents: Key Findings from PISA 2022

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Abstract

The present study explores the sources of validity evidence of the mathematics self-concept construct in Programme for International Student Assessment (PISA) 2022 with a focus on concurrent, discriminant, and predictive evidence. Mathematics self-concept was provided with a single index variable in PISA 2022. Using only the U.S. students who participated in PISA 2022, we included 4552 students. While we grounded our analyses in the *Standards* (2014), we aimed to evaluate whether the mathematics self-concept is related to mathematics motivation, distinct from mathematics self-efficacy measures and predicts mathematics achievement. Results revealed a positive association between mathematics self-concept and motivation ($\chi^2 = 86008.92$, $p < .001$, $\phi = 0.153$), suggesting concurrent evidence. Significant but negligible correlations with two separate mathematics self-efficacy measures ($r_{pb} = -0.078$ and -0.073 , $p < .001$) supported discriminant evidence. Multilevel modeling analysis demonstrated that mathematics self-concept significantly predicted mathematics performance ($\beta = 20.20$, $p < .001$), supporting predictive validity. Validity evidence for the mathematics self-concept scale in PISA 2022 could serve as a basis for research focused on enhancing student outcomes. Implications, limitations, and future directions are discussed.

Keywords

mathematics self-concept, large-scale assessment, psychometrics, validity

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Introduction

Academic self-concept is a key factor that reflects an individual's perception of their own abilities in an academic domain. These academic self-concepts have a critical role in shaping academic experiences and outcomes since they influence not only a student's engagement and motivation but also their overall academic achievement (Cooper et al., 2018; Jansen et al., 2021; Ramazan, Dai, et al., 2023; Steinberg et al., 2024). Domain-specific academic self-concepts, such as mathematics self-concept, have earned considerable attention due to their direct impact on adolescents' performance in specific subject areas (e.g., Dai et al., 2023; Gorges & Hollmann, 2019; Perinelli et al., 2022; Ramazan, Danielson, et al., 2023). For example, mathematics self-concept, which is typically defined as the self-perceived competence and confidence that an adolescent has in their ability to understand and perform mathematical tasks, is particularly important given the foundational role that mathematics has in education and its relevance to various career paths of adolescents (e.g., Ramazan, Danielson, et al., 2023; Sewasew et al., 2018). Previous work has shown that stronger mathematics self-concept is associated with mathematics-related positive outcomes (e.g., Hammoudi, 2019; Holenstein et al., 2021; Passiatore et al., 2023).

In the context of large-scale assessments like the Programme for International Student Assessment (PISA), understanding the validity of mathematics self-concept measures is critical to ensuring the accuracy and confidence in results and conclusions (AERA, APA & NCME, 2014; Hernández-Torrano & Courtney, 2021; Hopfenbeck et al., 2017). In particular, the PISA 2022 assessment offers a unique opportunity to explore the concurrent, discriminant, and predictive evidence of mathematics self-concept of adolescents on a global scale (Ding et al., 2022; Gustafsson, 2018; OECD, 2023) since it is the most recent and mathematics-focused PISA cycle (OECD, 2023; OECD, 2024). As it has been suggested by the Standards (AERA, APA & NCME, 2014), these aspects of validity evidence help ensure that a scale is not only theoretically sound but also practically useful and specific in its measurement. As well, these three aspects of validity evidence provide a solid foundation for validating a measure's relevance and effectiveness in its specific context (e.g., Clark & Watson, 2019; Strauss & Smith, 2009). That is, concurrent evidence supports alignment with other relevant measures, discriminant evidence suggests the separateness and specificity of different constructs, and predictive evidence helps establish real-world relevance and utility (Messick, 1989, 1995). It is because "...test developers and those selecting tests and interpreting test results need adequate knowledge of psychometric principles such as validity and reliability" (AERA, APA & NCME, 2014, p. 4). Examining concurrent, discriminant, and predictive validity evidence of a scale is crucial in psychometrics to ensure that the respective scale accurately and effectively measures what it is intended to measure since these sources of evidence are classified as evidence based on relations to other variables by the Standards (AERA, APA & NCME, 2014). By doing so, all three contribute to the overall robustness and credibility of the scale in both academic research and practical applications (e.g., Cook et al., 2015; Song et al., 2023; Valdivia & Li, 2021) as Standards indicated "*In many cases, the intended interpretation for a given use implies that the construct should be related to some other variables, and, as a result, analyses of the relationship of test scores to variables external to the test provide another important source of validity evidence*" (AERA, APA & NCME, 2014, p. 16).

Taken together, these forms of validity evidence provide a comprehensive understanding of how mathematics self-concept functions as a psychological construct, its relevance in educational settings, and its potential to inform educational practices and interventions aimed at improving student outcomes (e.g., AERA, APA & NCME, 2014; Assor & Connell, 1992; Danielson et al., 2023; Hudson et al., 2017; Valdivia & Li, 2021). Therefore, this research aimed to examine whether mathematics self-concept is related to mathematics motivation (i.e., concurrent evidence), how it is distinct from mathematics self-efficacy (i.e., discriminant evidence), and mathematics

self-concept is a predictor of mathematics performance (i.e., predictive evidence) in PISA 2022. On the one hand, previous literature has suggested that mathematics self-concept and mathematics motivation are usually found to have a strong relationship where a higher mathematics self-concept is associated with greater mathematics motivation (e.g., Hammoudi, 2019; Niemivirta et al., 2023; Skaalvik & Valås, 1999). This finding might seem counter-intuitive at first, but it can be understood in the context of self-enhancement theory (e.g., Calsyn & Kenny, 1977), which posits that positive self-perceptions in a subject can potentially lead to greater motivation to engage and perform well in that subject which could thereby reinforce and strengthen the self-concept. The literature has also been interested in showcasing how different self-concept and self-efficacy are from one another. Self-concept is a broad and overarching perception of one's identity and abilities which comprises various domains such as academic, social, and personal attributes. Self-efficacy, in contrast, specifically refers to an individual's belief in their ability to successfully perform tasks or achieve goals within a particular domain (Bong, 1998; Bong & Clark, 1999; Bong & Skaalvik, 2003). An extensive body of research (e.g., Ramazan, Danielson, et al. 2023; Sewasew et al., 2018; Skaalvik & Valås, 1999) has indicated that academic self-concept is predictive of academic achievement while it is also emphasized by the self-enhancement theory (e.g., Calsyn & Kenny, 1977) since self-concept is usually seen as a prerequisite for achievement. Thus, the following research questions have guided the present study:

RQ1: How strongly does the adolescents' mathematics self-concept in PISA 2022 relate to their mathematics motivation?

RQ2: To what extent does the adolescents' mathematics self-concept in PISA 2022 differentiate from their mathematics self-efficacy measures?

RQ3: How well does the adolescents' mathematics self-concept in PISA 2022 predict their mathematics achievement?

Methods

Participants

A total of 4552 fifteen-year-old and sixteen-year-old adolescents participated in PISA 2022 across the United States. The average age was $M = 15.83$ ($SD = 0.29$). Of the participants, 49.1% self-reported their sex as female ($N = 2235$) whereas 50.8% reported their sex as male ($N = 2312$). 0.1% chose not to report their sex ($N = 5$). English was the primary language of 76.76% of the participants ($N = 3494$) while 18.67% reported another language as their primary language ($N = 850$). 4.57% chose not to report their primary language ($N = 208$).

Variables

The main variable of interest was mathematics self-concept which was measured with a single item ("*Mathematics is easy for me*"), was calculated and used as an index variable and was coded with two response categories (0 = *No perception of mathematics as easier than other subjects*; 1 = *Perception of mathematics as easier than other subjects*). This index measures whether students perceive mathematics as easier than Test Language and Science. Examinees were assigned a "1" if they endorsed "*Mathematics is easy for me*" more strongly than both "*Test language is easy for me*" and "*Science is easy for me*"; otherwise, they received a "0." It reflects the relative ease of mathematics compared to the other subjects, rather than their absolute perception of mathematics

ease which is captured by the original item response. This implementation process has been determined and completed by PISA (OECD, 2024).

Mathematics motivation was also measured with a single item (“*I want to do well in my mathematics class*”), was calculated and used as an index variable and was coded with two response categories (0 = *Not more motivated to do well in mathematics than other subjects*; 1 = *More motivated to do well in mathematics than other subjects*). This index measures whether students are more motivated to succeed in mathematics than in Test Language and Science. Students received a “1” if they endorsed “*I want to do well in my mathematics class*” more strongly than both “*I want to do well in my test language class*” and “*I want to do well in my science class*”; otherwise, they received a “0.” This index reflects students’ relative motivation for mathematics, not their absolute motivation which is captured by the original item response. This implementation process has been determined and completed by PISA (OECD, 2024).

Mathematics self-efficacy in PISA 2022 was assessed with two separate composite variables: (1) mathematics self-efficacy: formal and applied mathematics, and (2) mathematics self-efficacy: mathematical reasoning and 21st century mathematics. First, mathematics self-efficacy: formal and applied mathematics ($M = -0.44$; $SD = 1.15$) was measured with nine items (e.g., “*How confident in math tasks: Working out from a [train timetable] how long it would take to get from one place to another*”) and with four response categories (1 = *not at all confident* to 4 = *very confident*). Second, mathematics self-efficacy: mathematical reasoning and 21st century mathematics ($M = 0.08$; $SD = 1.00$) was measured with 10 items (e.g., “*How confident in math tasks: Extracting mathematical information from diagrams, graphs, or simulations*”) and with four response categories (1 = *not at all confident* to 4 = *very confident*). These two composite variables were created by using item response theory scaling (OECD, 2024).

Mathematics performance focused on participants’ ability to deduce from what they know and implement their knowledge in different contexts (OECD, 2023; OECD, 2024). Mathematics performance was included in the data as 10 sets of plausible values (Means ranged from $M = 461.87$; $SD = 95.13$ to $M = 463.62$; $SD = 95.07$) due to booklet assessment designs in PISA. Since PISA assessments cannot capture all mathematics outcomes for every student due to time constraints and the risk of student fatigue, missing data in mathematics achievement scores occurred by design as different students received different assessment booklets. To address this, PISA implemented an imputation strategy to generate plausible values as proxies of achievement scores. In other words, plausible values are estimated scores derived through imputation based on available data. Since plausible values are only estimations, it is a standard practice to use multiple sets of plausible values (usually 5 to 10 sets) in secondary analysis for the results to be more accurate. This implementation process has been determined and completed by PISA (OECD, 2024). In analysis using large-scale assessment data collected from a booklet design, such as PISA, it is standard practice to use all sets of plausible values (Rutkowski et al., 2010).

All of the variable labels, the items that were used to create the variables and indices were provided by the PISA. More information regarding the variables of interest in the present study and plausible values can be found in the PISA 2022 Technical Report (OECD, 2024).

Data Analysis

The data used and analyzed for the present study were retrieved from the publicly available PISA 2022 data repository (see <https://www.oecd.org/en/data/datasets/pisa-2022-database.html>).

The proportions of the missing data ranged from 11.9% to 13.5% across the variables of the present study. All missing data were reported as “no response” by PISA 2022, and systematic missing-by-design does not apply to background questionnaire data. Therefore, prior to the

analysis, the missing data were imputed by using a fully conditional specification method with a single imputation and 10 iterations (Enders et al., 2018).

As aforementioned, the present study examined validity evidence based on relations to other variables according to the Standards (AERA, APA & NCME, 2014). **Concurrent evidence** was examined with a chi-squared (χ^2) test between mathematics self-concept and mathematics motivation along with phi coefficient (ϕ) and Cramer's V as the effect size measures. **Discriminant evidence** was assessed with a point biserial correlation (r_{pb}) between (1) mathematics self-concept and mathematics self-efficacy: formal and applied mathematics and between (2) mathematics self-concept and mathematics self-efficacy: mathematical reasoning and 21st century. The correlation coefficients were also interpreted as the respective effect sizes. Finally, **predictive evidence** was tested with a multilevel modeling approach where mathematics self-concept was used to predict mathematics performance. The between-cluster variance in multilevel modeling analysis was taken into account by employing the replication weights (OECD, 2024). In the models, the students were nested within the schools (e.g., OECD, 2009). Explicitly, the null or unconditional model (i.e., the model with no predictors) comprised the mathematics performance randomly varying across the schools (i.e., random intercepts). The conditional or random intercept model reported the parameters where mathematics self-concept was used to predict mathematics performance.

Sampling weights were used in all analyses to obtain precise parameters (OECD, 2024). The analyses were performed in the International Database [IDB] Analyzer v5.0.32 (IEA, 2022) and SPSS v25 (IBM, 2017).

Results

Chi-squared test results (Table 1) revealed a statistically significant association between the adolescents' mathematics self-concept and their mathematics motivation ($\chi^2 = 86008.920$, $df = 1$, $p < .001$, $\phi = 0.153$, Cramer's $V = 0.153$), suggesting a moderate-to-strong effect size (Akoglu, 2018). This association indicates that the observed relationship between mathematics self-concept and mathematics motivation is unlikely to be attributable to random variation under the null hypothesis. The results showed that higher levels of mathematics self-concept were associated with higher levels of mathematics motivation. This supported the concurrent validity evidence of the mathematics self-concept scale, which could indicate that it effectively aligned with and was indicative of related constructs within the PISA 2022 framework.

Subsequently, point biserial correlation results (Table 2) showed a significant, negative but negligible correlation between the adolescents' mathematics self-concept and their mathematics self-efficacy: formal and applied mathematics ($r_{pb} = -0.078$, $p < .001$). Similarly, there was also a significant, negative but negligible correlation between the adolescents' mathematics self-concept and their mathematics self-efficacy: mathematical reasoning and 21st century ($r_{pb} = -0.073$, $p < .001$). These negative and significant correlations with negligible effect sizes are practically 0 (zero), suggesting that mathematics self-concept was meaningfully distinct from the constructs measured by (1) mathematics self-efficacy: formal and applied mathematics and (2) mathematics self-efficacy: mathematical reasoning and 21st century. Specifically, the negligible correlation

Table 1. Chi-Squared Test Results.

χ^2	df	p	ϕ	Cramer's V
86008.920	1	<0.001	0.153	0.153

coefficients indicated that mathematics self-concept, and mathematics self-efficacy: formal and applied mathematics and mathematics self-efficacy: mathematical reasoning and 21st century were not substantially overlapping constructs where it supported the discriminant validity evidence of mathematics self-concept. This differentiation is crucial for suggesting that mathematics self-concept measures a unique aspect of students’ perceptions of their mathematical abilities which is separate from their self-efficacy beliefs.

Finally, the multilevel modeling analysis results (Table 3) established that the adolescents’ mathematics self-concept was a significant and positive predictor of their mathematics performance ($\beta = 20.20$, $se = 4.28$, $t = 4.72$, $p < .001$). This significant and positive relationship showed that students who perceived themselves more favorably in mathematics outperformed their counterpart peers by an average of 20.20 points on mathematics assessments. Thus, the predictive validity evidence of the mathematics self-concept scale was supported, which also highlighted its effectiveness in predicting mathematics performance based on students’ mathematics self-concept.

Discussion

The findings of the present study provided support for the concurrent, discriminant, and predictive evidence of the mathematics self-concept scale within the context of PISA 2022 as validity evidence based on relations to other variables (AERA, APA & NCME, 2014). These results have significant implications for educational research, practice and policy-making in mathematics education of adolescents. In what follows, we discussed the findings of the present study with their relevance to the previous research.

The significant association between adolescents’ mathematics self-concept and mathematics motivation supported the concurrent validity evidence of the mathematics self-concept scale. Consistent with previous research (e.g., Hammoudi, 2019; Niemivirta et al., 2023), the positive relationship between these constructs indicated that students who perceive themselves as more competent in mathematics were also more motivated to succeed in mathematics. This indicated the potential importance of fostering positive self-concepts in mathematics to potentially enhance motivation which could in turn improve student outcomes (Skaalvik & Valås, 1999).

The significant but negligible correlations between adolescents’ mathematics self-concept and two mathematics self-efficacy measures supported the discriminant validity evidence of the mathematics self-concept scale. The findings highlighted the distinct nature of mathematics self-concept and mathematics self-efficacy as the negligible correlations suggested that these

Table 2. Point Biserial Correlation Results.

Variables	Mathematics Self-Concept
Mathematics self-concept	-
Mathematics self-efficacy: Formal and applied mathematics	−0.078***
Mathematics self-efficacy: Mathematical reasoning and 21st century	−0.073***

Note. *** = $p < .001$.

Table 3. Multilevel Modeling Analysis Results to Predict Mathematics Performance.

Predictor	Standardized β	Unstandardized β	se	t	p
Mathematics self-concept	0.08	20.20	4.28	4.72	<0.001

constructs did not overlap substantially. This finding aligns with prior research distinguishing self-concept as a broader perception of one's abilities whereas self-efficacy refers to the confidence in successfully performing specific tasks (e.g., Bong & Clark, 1999; Bong & Skaalvik, 2003). The separation of these constructs may be crucial for educational research and practice. Understanding this distinction could allow educators and researchers to develop more nuanced educational practices that address different aspects of students' perceptions of their abilities in mathematics.

The results also indicated that adolescents' mathematics self-concept is a significant predictor of their mathematics performance. This supports the predictive validity evidence of the mathematics self-concept scale and was consistent with the self-enhancement theory which posits that higher self-concept could lead to better academic achievement (e.g., Calsyn & Kenny, 1977; Sewasew et al., 2018). The results suggested that students with a stronger self-concept in mathematics tended to perform significantly better on mathematics assessments. As such, to put this into perspective, we found that a positive mathematics self-concept contributed to 20.20-point increase in mathematics performance, which potentially could shift a student's performance from the median to a higher percentile given the mean range of $M = 461.87$ ($SD = 95.13$) to $M = 463.62$ ($SD = 95.07$). These findings point to the importance of fostering a positive self-concept in mathematics to enhance students' academic performance. This predictive relationship has practical implications for educators since interventions aimed at improving mathematics self-concept could potentially lead to improvements in students' academic achievement. We discuss these implications next.

Implications

Given the fact that mathematics self-concept is related to motivation and achievement, educators could consider strategies to enhance adolescents' self-perceptions in mathematics. However, the directionality of this relationship remains uncertain; it is equally plausible that higher achievement and motivation contribute to more positive self-concept. Therefore, holistic approaches could be taken into account to address multiple factors that are related to students' academic experiences. Additionally, educators should be mindful of the distinction between self-concept and self-efficacy when designing interventions to ensure that both perceptions of competence and specific task-related confidence are addressed. The significant predictive relationship between mathematics self-concept and performance may suggest that self-concept measures could be valuable in predictive models of academic achievement. Applied researchers could consider incorporating self-concept into machine learning or other statistical models to predict student outcomes more accurately grounded in the Standards (AERA, APA & NCME, 2014) to adhere to principles of validity, reliability, fairness, and transparency in educational measurement. Incorporating mathematics self-concept into predictive models should be theoretically and empirically justified in order to ensure consistent and unbiased measurement across diverse student populations. Models should be interpretable and ethically designed to support equitable educational outcomes.

Limitations and Future Directions

While this study provided evidence for the validity of the mathematics self-concept scale, it is not without limitations. First, the use of single index item measures for some variables (e.g., mathematics motivation) may limit the depth of the data collected. Future studies should consider employing multi-item scales to capture a more nuanced understanding of these constructs. Additionally, the present study focused on a specific cycle of the PISA 2022 dataset, which may limit the generalizability of the findings to other populations or educational contexts. Future

research should explore the validity evidence of the mathematics self-concept scale across different cultural contexts, populations and educational systems.

Conclusion

In conclusion, grounded in the Standards (AERA, APA & NCME, 2014), this study provides support for the concurrent, discriminant, and predictive validity evidence of the mathematics self-concept scale in PISA 2022 (i.e., evidence based on relations to other variables). These findings highlight the importance of mathematics self-concept in relation to mathematics motivation and mathematics performance and to what extent it is distinct from mathematics self-efficacy measures.

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Authors' Contributions

OR led and contributed to study conceptualization, data management and analysis, methods and results sections, interpretation of the results, writing original draft, and contributed to other components of the present manuscript. SD contributed to data management and analysis, methods and results sections, interpretation of the results, writing original draft, and contributed to other components of the present manuscript. RWD contributed to the conceptualization and design of the present study, interpretation of the results, writing original draft, and contributed to other components of the present manuscript. All authors read and approved the final manuscript.

Ethical statement

The present study has been conducted according to the American Psychological Association's ethical standards.

Ethics Approval

Not applicable

Consent to Participate

Not applicable

Consent for Publication

Not applicable

Conflict of Interest

The authors declare that they have no competing interests.

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Data Availability Statement

The datasets generated and/or analyzed during the current study are available in the Programme for International Student Assessment 2022 data repository, <https://www.oecd.org/en/data/datasets/pisa-2022-database.html>.

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